

EXPERIMENT: 10

5G NETWORK SLICING ENVIRONMENT USING MATLAB

Objective – 1: Analyze the Number of Network Slices

MATLAB Code:

```
clc;

clear;

close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1]; % One instance of each network slice

figure

bar(count)

grid on

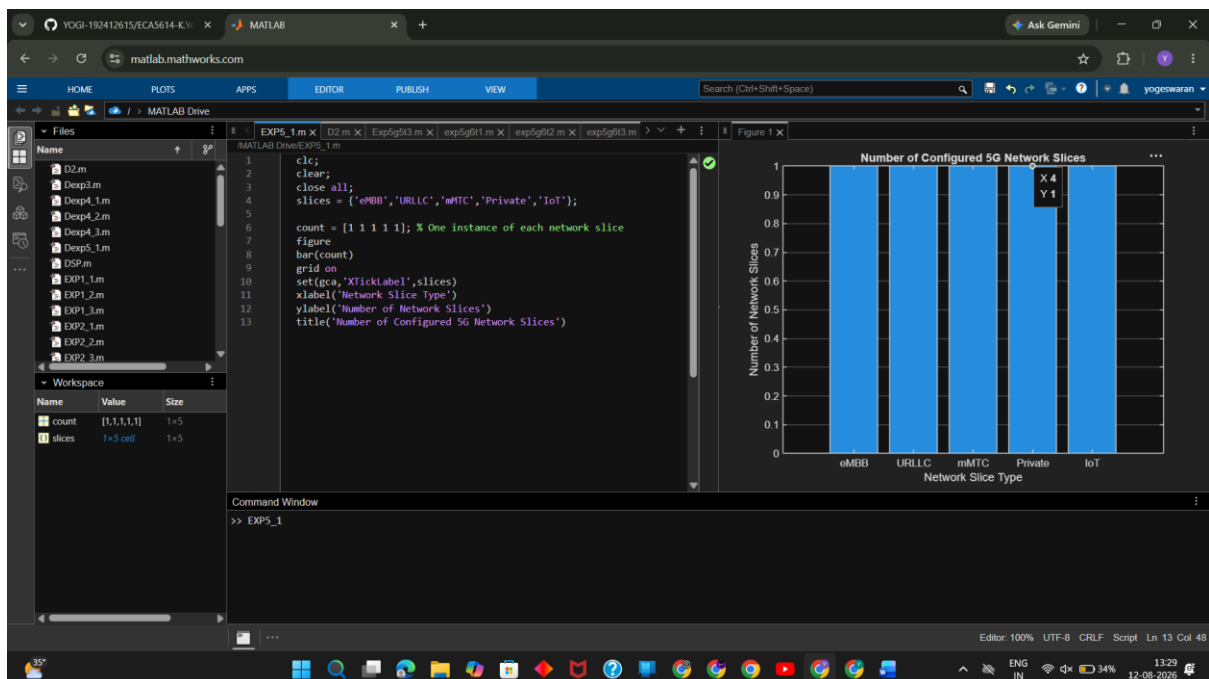
set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

title('Number of Configured 5G Network Slices')
```

OUTPUT



Objective – 2: Compare Slice Utilization (%)

Matlab Code :

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

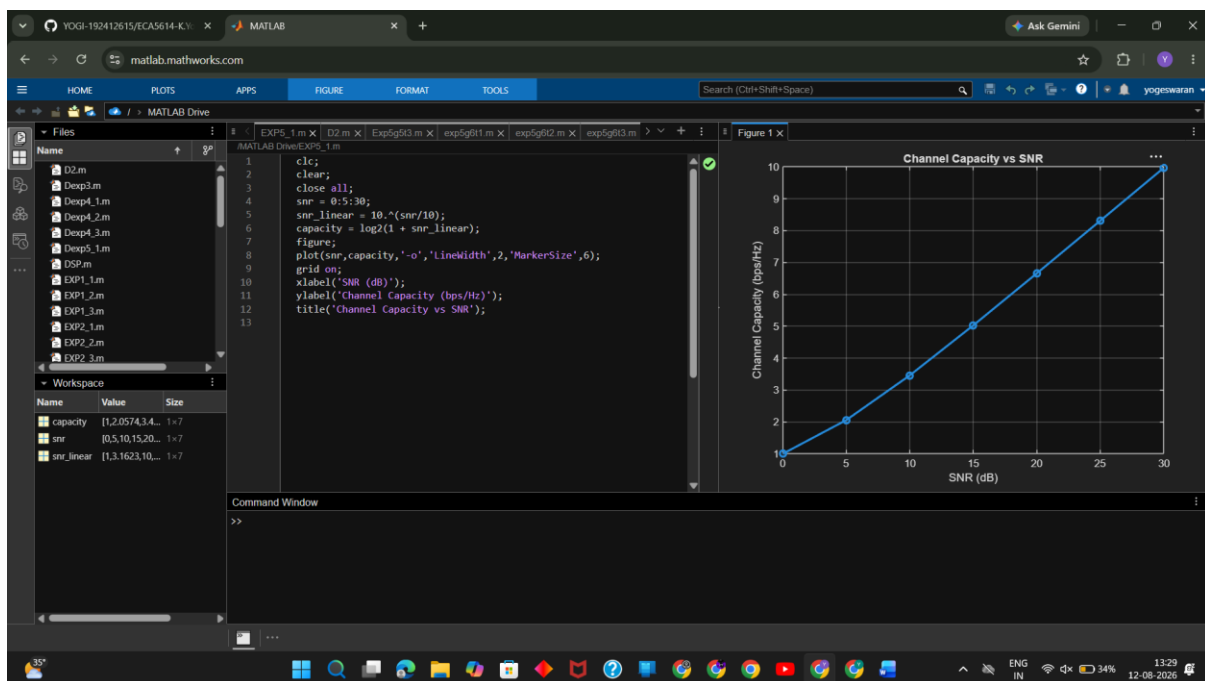
grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```

OUTPUT



Objective – 3: Analyze End-to-End Latency (ms)

MATLAB Code:

```
clc;

clear;

close all;
```

```
antennas = [4 8 16 32 64]; % Number of Antenna Elements
```

```
gain = 10*log10(antennas); % Antenna Gain (dB)
```

```
figure
```

```
bar(antennas, gain)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Antenna Gain (dB)')
```

```
title('Antenna Gain for Different Antenna Array Sizes')
```

OUTPUT

